BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade)

Syllabus: Unit 1 — Physical Quantities & Measurement | Unit 2 — Kinematics | Unit 3 — Dynamics-I | Unit 4 — Dynamics-II | Unit 5 — Pressure & Deformation in Solids | Unit 6 — Work & Energy (till 6.4) | Unit 9 — Nature of Science

ANSWER KEY (Version 1)

Total Marks: 65	**Time Allowed: 2 hours 30 minutes**	**Version: 1**

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Q.1: A — 3 1
Explanation: Leading zeros before the first nonzero digit are never **significant**. In 0.00650, the digits 6, 5 and 0 are significant (a trailing zero after the decimal point, following nonzero digits, is significant). Hence it has **3 significant figures**.

Q.2: B — 0.05 mm 1
Explanation: Least count = smallest main scale division ÷ number of vernier divisions = 1 mm ÷ 20 = **0.05 mm**.

Q.3: C — Zero 1
Explanation: Displacement is the shortest straight-line distance between the initial and final positions. Since the athlete returns to her exact starting point, her displacement is zero even though the distance covered (400 m) is not.

Q.4: D — 3 m/s² 1
Explanation: $a = (v_f - v_i)/t = (22 - 4)/6 = 18/6 =$ **3 m/s²**.

Q.5: C — Newton's 3rd law 1
Explanation: Newton's **third law** states that for every action there is an equal and opposite reaction. The swimmer's hands push the water backward (action) and the water pushes her forward (reaction) — these forces act on **different bodies** and do not cancel.

Q.6: C — 1800 kg·m/s 1
Explanation: Momentum p = mv = 300 kg × 6 m/s = **1800 kg·m/s**.

Q.7: B — Net torque 1
Explanation: The **second condition of equilibrium** (rotational equilibrium) requires that the sum of all torques (moments) acting on the body about any pivot must be zero. The first condition (net force zero) governs translational equilibrium instead.

Q.8: A — 6.4 N 1
Explanation: $F_c = mv^2/r = (0.2 \text{ kg} \times 4^2)/0.5 \text{ m} = (0.2 \times 16)/0.5 = 3.2/0.5 =$ **6.4 N**.

Q.9: D — 200 N/m 1
Explanation: By **Hooke's law**, k = F/x = 16 N/0.08 m = **200 N/m**.

Q.10: C — 760 mmHg 1
Explanation: The standard **atmospheric pressure** at sea level is 1 atm = 760 mmHg = 1.013×10^5 Pa. This is the height to which mercury rises in a barometer tube at sea level.

Q.11: D — 240 J 1
Explanation: Gravitational potential energy $E_p = mgh = 3 \text{ kg} \times 10 \text{ m/s}^2 \times 8 \text{ m} =$ **240 J**.

Q.12: B — Particle physics 1
Explanation: Particle physics (or high-energy physics) is the study of fundamental particles and the forces that create matter and radiation. Astrophysics studies stars and celestial bodies; acoustics studies sound; geophysics studies Earth's structure.

SECTION B — Short Answer Questions (11 × 3 = 33 Marks)

2(i) Scientific Notation & SI Prefixes 3

(a) 0.00072 m = $\mathbf{7.2 \times 10^{-4}}$ **m** 1

(b) 8,900,000 m = $\mathbf{8.9 \times 10^{6}}$ **m** 1

Using an appropriate prefix: 7.2×10^{-4} m = 0.72×10^{-3} m = **0.72 mm** (milli). 1

OR

SI Prefixes & Conversion 3

10^2 = **hecto (h)**, 10^{-2} = **centi (c)**, 10^{12} = **tera (T)**. 2

8 km = 8×10^3 m = **8000 m**. 1

2(ii) Least Count of Vernier Caliper 3

Least count is the minimum length that can be measured accurately with a vernier caliper. 1

Least count = **smallest main scale division ÷ total vernier divisions**. 1

Least count = 1 mm ÷ 25 = **0.04 mm** (= 0.004 cm). 1

OR

Rounding Off to 3 Significant Figures 3

(a) 45.268 → **45.3** 1

(b) 0.005672 → **0.00567** 1

(c) 9,432 → **9,430** 1

2(iii) Distance & Displacement 3

Total distance = 6 km + 8 km = **14 km**. 1

Since the two legs of the journey are perpendicular (north then east), displacement is found using Pythagoras' theorem: $d = \sqrt{6^2 + 8^2} = \sqrt{36+64} = \sqrt{100}$. 1

Displacement = **10 km** (directed north-east of the starting point). 1

OR

Average vs Instantaneous Speed 3

Average speed is the total distance covered divided by total time taken for the whole journey. 1

Instantaneous speed is the speed of an object at any one specific instant of time (e.g. the reading on a speedometer at that moment). 1

Average speed of bus = 210 km / 3 h = **70 km/h**. 1

2(iv) Acceleration & Distance Covered 3

(a) Acceleration $a = (v_f - v_i)/t = (25 - 5)/8 = 20/8 =$ **2.5 m/s²**. 1

(b) Average velocity = $(v_i + v_f)/2 = (5+25)/2 = 15$ m/s. 1

Distance = average velocity × time = 15 m/s × 8 s = **120 m**. 1

OR

Free Fall — Final Velocity & Height 3

(a) $v = g \times t = 10\ \text{m/s}^2 \times 4\ \text{s} =$ **40 m/s**. 1

(b) $h = \frac{1}{2} g t^2 = \frac{1}{2} \times 10 \times 4^2 = \frac{1}{2} \times 10 \times 16 =$ **80 m**. 1

Both results follow from the equations of motion applied to a freely falling body starting from rest. 1

2(v) Newton's Second Law — Acceleration 3

For 4 kg trolley: $a = F/m = 18$ N / 4 kg = **4.5 m/s²**. 1

For 12 kg trolley: $a = F/m = 18$ N / 12 kg = **1.5 m/s²**. 1

For the same net force, **acceleration is inversely proportional to mass** — the heavier trolley accelerates three times more slowly. 1

OR

Inertia & Jumping Off a Moving Bus 3

Inertia is the natural tendency of a body to resist any change in its state of rest or uniform motion. 1

While inside a moving bus, your whole body moves forward with the bus's velocity. 1

If you jump off, your feet stop suddenly on contact with the ground (due to friction), but the rest of your body continues moving forward because of inertia — causing you to fall and get injured. This is why jumping off a moving bus is dangerous. 1

2(vi) Impulse & Average Force 3

(a) Impulse = change in momentum = $m(v_f - v_i)$ = 0.2 kg × (0 − 15 m/s) = −3 kg·m/s, magnitude **3 N·s**. 1

(b) Average force $F = \Delta p/\Delta t = -3/0.1 = -30$ N, magnitude **30 N**. 1

The negative sign shows the force acts opposite to the ball's original direction of motion (bringing it to rest). 1

OR

Conservation of Momentum — Recoil 3

Total initial momentum of the (girl + ball) isolated system = 0 (both at rest). 1

By conservation of momentum: $0 = m_{girl}v_{girl} + m_{ball}v_{ball} = 40v_{girl} + (4 \times 5)$. 1

$v_{girl} = -20/40 =$ **−0.5 m/s** — the girl recoils at 0.5 m/s in the direction opposite to the ball. 1

2(vii) Torque of a Spanner 3

Torque $\tau = F \times d$ = 60 N × 0.30 m = **18 N·m**. 1

For the shorter spanner (d = 0.15 m) to produce the same torque: $F = \tau/d = 18/0.15 =$ **120 N**. 1

A shorter spanner needs a **larger force** to produce the same torque, since torque = force × moment arm, and the moment arm is smaller. 1

OR

Principle of Moments — Seesaw 3

By the **principle of moments**: anticlockwise moment = clockwise moment, i.e. $W_{Ayesha} \times d_{Ayesha} = W_{Bilal} \times d_{Bilal}$. 1

480 N × 0.25 m = 400 N × $d_{Bilal} \Rightarrow d_{Bilal} = 120/400$. 1

d_{Bilal} = **0.3 m** from the pivot. 1

2(viii) Centripetal Force 3

$F_c = mv^2/r$ = (0.4 kg × 5²)/0.8 m = (0.4 × 25)/0.8 = 10/0.8. 1

F_c = **12.5 N**. 1

This force is directed **towards the centre** of the circular path, keeping the ball from moving off in a straight line. 1

OR

Average Orbital Speed 3

$v_{avg} = 2\pi r/T$ = (2 × 3.14 × 7,000,000 m)/6000 s. 1

v_{avg} = 43,982,000/6000 ≈ 7330 m/s. 1

$v_{avg} \approx$ **7.33 km/s**. 1

2(ix) Spring Constant (Hooke's Law) 3

Force stretching spring = weight = mg = 3 kg × 10 m/s² = 30 N. 1

By Hooke's law, k = F/x = 30 N/0.06 m. 1

k = **500 N/m**. 1

OR

Pressure Exerted by a Box 3

Pressure P = F/A = 240 N/0.3 m². 1

P = **800 Pa** (= 800 N/m²). 1

Pressure is force applied per unit area; the same force spread over a larger area produces smaller pressure and vice versa. 1

2(x) Kinetic Energy 3

KE_1 = ½mv² = ½ × 2 × 12² = ½ × 2 × 144 = **144 J**. 1

If speed doubles to 24 m/s: KE_2 = ½ × 2 × 24² = ½ × 2 × 576 = **576 J**. 1

Since KE ∝ v², doubling the speed makes kinetic energy **four times** greater (576 J = 4 × 144 J). 1

OR

Energy Conversion & Conservation 3

A body raised to a height has **gravitational potential energy** E_p = mgh. As it falls, this potential energy is progressively converted into **kinetic energy**. 1

Example: a ball dropped from a shelf loses height (and PE) while gaining speed (and KE) as it falls, until just before hitting the ground almost all of its PE has become KE. 1

Law of conservation of energy: energy can neither be created nor destroyed, only converted from one form to another; the total energy of an isolated system remains constant. 1

2(xi) Newton's Laws & Einstein's Relativity 3

Newton's laws of motion accurately describe motion at everyday (normal) speeds and were treated as universally true for over two centuries. 1

However, these laws **fail for objects moving at speeds approaching the speed of light**. Albert Einstein resolved this with his **theory of relativity**, which gives the same results as Newton's laws at low speeds but remains accurate even near light speed. 1

This shows that scientific theories are **provisional**: they are refined or extended (not simply discarded) as new evidence and more precise experiments become available. 1

OR

Fields of Physics 3

Biophysics: studies characteristics and systems of the living body, e.g. fluid dynamics of blood flow. 1

Astrophysics: studies the physical nature of stars and celestial bodies, e.g. the formation of stars. 1

Acoustics: studies the production, transmission and reception of sound, e.g. designing concert hall acoustics. (Any three of the four count for full marks, one mark each.) 1

SECTION C — Long Questions (4 × 5 = 20 Marks)

3(i) Vernier Caliper: Construction, Working & Reading 5

Construction & working: 2

A vernier caliper consists of a **main scale** (marked in mm, with a fixed jaw) and a sliding **vernier scale** with jaws that can move along the main scale. It also has upper jaws for measuring internal diameters and a depth probe. The object to be measured is gripped between the jaws; the vernier scale then slides to a position where its zero mark gives the main scale reading, and the vernier division that best coincides with a main scale division gives the fractional part of the reading.

Reading calculation: 1

Total reading = Main Scale Reading (MSR) + (Vernier Scale Reading (VSR) × Least Count).

Calculation: 2

Total reading = 3.2 cm + (6 × 0.01 cm) = 3.2 + 0.06 = **3.26 cm**.

OR

Screw Gauge: Definition, Pitch, Least Count & Diameter 5

Definition: 1 A screw gauge is a device used to measure a fraction of the smallest division on a linear scale by rotating a circular scale over it — it measures lengths even smaller than a vernier caliper.

Pitch & least count: 1 **Pitch** is the distance moved by the circular scale along the main scale in one complete rotation. **Least count** = pitch ÷ number of divisions on the circular scale = 1 mm/100 = **0.01 mm**.

Diameter calculation: 3 Diameter = Main Scale Reading + (Circular Scale Reading × Least Count) = 4 mm + (32 × 0.01 mm) = 4 + 0.32 = **4.32 mm**.

3(ii) Newton's Second Law & Cricket Ball Impact 5

Newton's second law: 1 The acceleration produced by a net force on an object is directly proportional to the force and inversely proportional to its mass, in the direction of the force: $F_{net} = ma$. In terms of momentum, $F_{net} = \Delta p/\Delta t$ (the time rate of change of momentum).

(a) Change in momentum: 2

$\Delta p = m(v_f - v_i)$ = 0.15 kg × (−15 − 25) m/s = 0.15 × (−40) = −6 kg·m/s, magnitude **6 N·s** (rebound taken as negative direction).

(b) Average force: 2

$F = \Delta p/\Delta t = -6/0.01 = -600$ N, magnitude **600 N**. This very large force arises because the ball's momentum reverses direction in a very short contact time.

OR

Newton's Three Laws & Collision 5

Newton's three laws with examples: 2

- **First law (inertia):** a book on a table stays at rest until a force moves it.
- **Second law:** pushing a shopping trolley harder produces greater acceleration.
- **Third law:** a swimmer pushes water back and is pushed forward in reaction.

Conservation of momentum setup: 1 $m_1v_1 + m_2v_2 = (m_1+m_2)v$, since the car is initially at rest ($v_2=0$).

Calculation: 2 (2000 × 15) + (800 × 0) = (2000+800)v ⇒ 30,000 = 2800v ⇒ v = **10.71 m/s** (common velocity after the collision).

3(iii) Conditions of Equilibrium & Metre Rule 5

Two conditions of equilibrium: 1

First condition: the vector sum of all forces acting on the body must be zero (translational equilibrium).
Second condition: the sum of all torques (moments) about any pivot must be zero (rotational equilibrium).

Setting up the moment equation: 2

Moment arm of the 8 N weight from the pivot (50 cm mark) = 50 − 20 = 30 cm = 0.3 m. Its (anticlockwise) moment = 8 N × 0.3 m = 2.4 N·m.

Finding the position of the 5 N weight: 2

By the principle of moments, the 5 N weight's clockwise moment must equal 2.4 N·m: d = 2.4/5 = 0.48 m = 48 cm from the pivot, on the other side. Its position on the rule = 50 + 48 = **98 cm mark**.

OR

Centripetal Force & Orbital Motion — Satellite 5

Definitions: 1 **Centripetal force** is the force that pulls an object out of its straight-line path and into a circular path, directed towards the centre. **Orbital motion** is the regular, repeating circular (or near-circular) path that a satellite follows around a larger body due to gravity acting as the centripetal force.

(a) Average orbital speed: 2 v = 2πr/T = (2 × 3.14 × 8,000,000)/7200 = 50,265,000/7200 ≈ **6981 m/s** (≈ 6.98 km/s).

(b) Centripetal force: 2 F_c = mv²/r = (500 × 6981²)/8,000,000 = (500 × 48,738,000)/8,000,000 ≈ **3046 N** (≈ 3.05 kN).

3(iv) Pascal's Principle & Hydraulic Lift 5

Pascal's principle & hydraulic lift: 2

Pascal's principle: an external pressure applied to an enclosed fluid is transmitted unchanged to every point within the fluid. In a **hydraulic lift**, a small force applied to a narrow piston (small area A_1) creates a pressure that is transmitted equally to a wider piston (large area A_2), producing a much larger force there: $F_2/A_2 = F_1/A_1$, so $F_2 = (A_2/A_1)F_1$.

Calculation: 3

Weight of car F_2 = mg = 1200 kg × 10 m/s² = 12,000 N.

From Pascal's principle, $F_1 = (A_1/A_2) \times F_2$ = (0.004/0.8) × 12,000 = 0.005 × 12,000 = **60 N**.

OR

Deriving P = ρgh & Water Tank Pressure 5

Derivation: 3

Consider a column of liquid of height h, base area A and density ρ. The force on the base equals the weight of the liquid column: F = W = mg. Since density ρ = m/V and V = Ah, mass m = ρAh. So F = ρAhg. Since pressure P = F/A, substituting gives P = (ρAhg)/A, i.e. **P = ρgh**.

Calculation: 2

P = ρgh = 1000 kg/m³ × 10 m/s² × 6 m = **60,000 Pa = 60 kPa**.